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Centre Number

Candidate Number

Candidate Name _____

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UNIVERSITY OF CAMBRIDGE LOCAL EXAMINATIONS SYNDICATE

**Joint Examination for the Higher School Certificate
and General Certificate of Education Advanced Level**

PHYSICS

9243/4

PAPER 4 Practical Test (Alternative A)

Monday

23 OCTOBER 2000

3 hours

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Electronic calculator

Graph paper

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page and on any separate answer paper used.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

In Questions 1 and 2, you are expected to record all your observations as soon as these observations are made, and to plan the presentation of the records so that it is not necessary to make a fair copy of them. The working of the answers is to be handed in. Marks are mainly given for a clear record of the observations actually made, for their suitability and accuracy, and for the use made of them.

Routine precautions and theory are **not** wanted in Questions 1 and 2. You should, however, record any **special** precautions you have taken so as to aid accuracy.

At the end of the examination, fasten any separate answer paper used securely to the question paper.

INFORMATION FOR CANDIDATES

Questions 1 and 2 carry the same number of marks.

You will be allowed one hour only with the apparatus for each of Questions 1 and 2.

Questions 3 and 4 together carry about the same number of marks as either Question 1 or Question 2.

Squared paper and Mathematical tables are available.

Additional answer paper and graphs should be submitted **only** if it becomes **necessary** to do so.

You are advised to spend approximately one hour on each of Questions 1 and 2 and 30 minutes on each of Questions 3 and 4.

You are reminded of the need for good English and clear presentation in your answers.

FOR EXAMINER'S USE

1	
2	
3	
4	
TOTAL	

This question paper consists of 11 printed pages, 1 blank page and 4 lined pages.

It is recommended that you spend about 60 minutes on this question.

- 1 In this experiment, you will investigate an aspect of a phase relationship between two oscillating pendulums of differing natural frequency.

A pendulum (labelled A) of length 50.0 cm has been suspended from a stand. You should not alter the length of this pendulum during the course of the experiment.

- (a) In front of pendulum A, suspend a second pendulum (labelled B) using the split cork so that the points of suspension of both pendulums are at the same height. The separation of these points should be about 10 cm and the length l_B of pendulum B should be 45.0 cm. See Fig. 1.1.

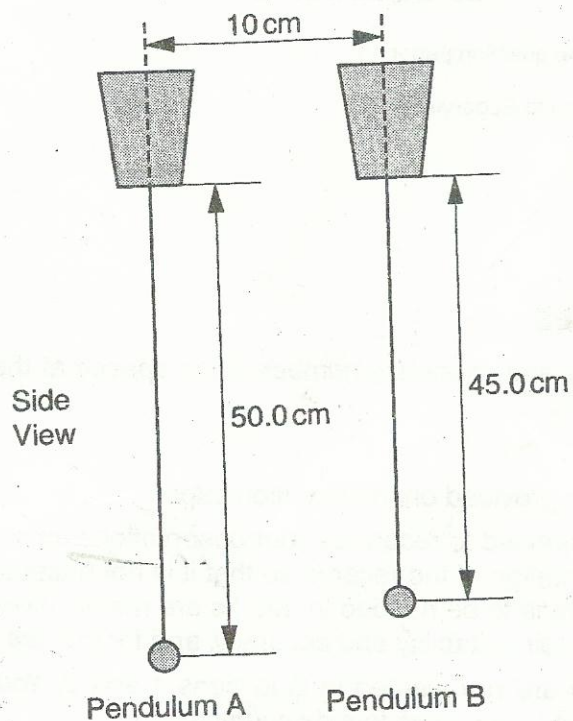


Fig. 1.1

- (b) (i) Gently displace ~~the~~ ^{the bottom} of the pendulum bobs by the same amount in a direction perpendicular to the plane containing the pendulums (see Fig. 1.2). Release both bobs at the same time. Note that the two pendulum bobs, which are initially in phase (i.e. are at the same point in their paths and are moving in the same direction), become out of phase before returning to a situation where they are in phase again. The time taken for this to occur is τ .

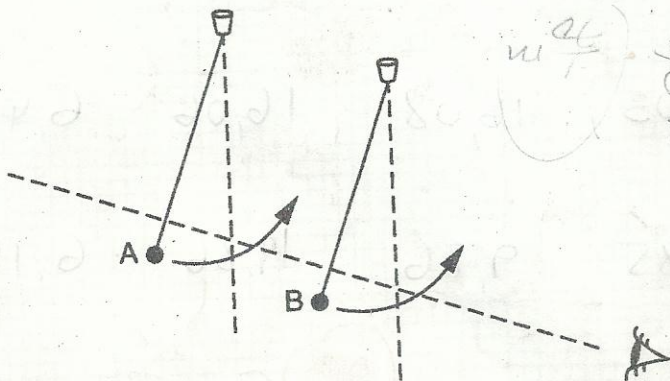


Fig. 1.2

- (ii) Measure and record the value of τ and the value of l_B .
- (iii) Estimate the uncertainty in this value of τ and state one way in which this uncertainty may be reduced.
- (iv) Change the value of l_B and repeat (ii) until you have six sets of readings for τ and l_B . l_B should be in the range $25 \text{ cm} < l_B < 50 \text{ cm}$. Include values of $\sqrt{l_B}$ and $\frac{\sqrt{l_B}}{\tau}$ in your table of results.
- (c) l_B and τ are related by the equation

$$\frac{\sqrt{l_B}}{\tau} = p\sqrt{l_B} + q$$

where p and q are constants.

- (i) Plot a graph of $\frac{\sqrt{l_B}}{\tau}$ (y-axis) against $\sqrt{l_B}$ (x-axis).
- (ii) Determine the gradient and the y-intercept of the graph.
- (iii) State the values of the constants p and q . You should include appropriate units in each case.
- (d) Use the results of your experiment to calculate how many times the pendulums, initially in phase, would go out of phase and back into phase again during a time interval of 20 minutes when $l_B = 5.0 \text{ cm}$.

It is recommended that you spend about 60 minutes on this question.

- 2 In this experiment you will investigate how the resistance of a loop of wire varies with the distance between two points on the wire.

You are provided with a loop of constantan wire. The loop has been taped to a metre rule and attached to a connecting lead. You should not disturb this connection during the experiment.

- (a) Construct the circuit shown in Fig. 2.1.

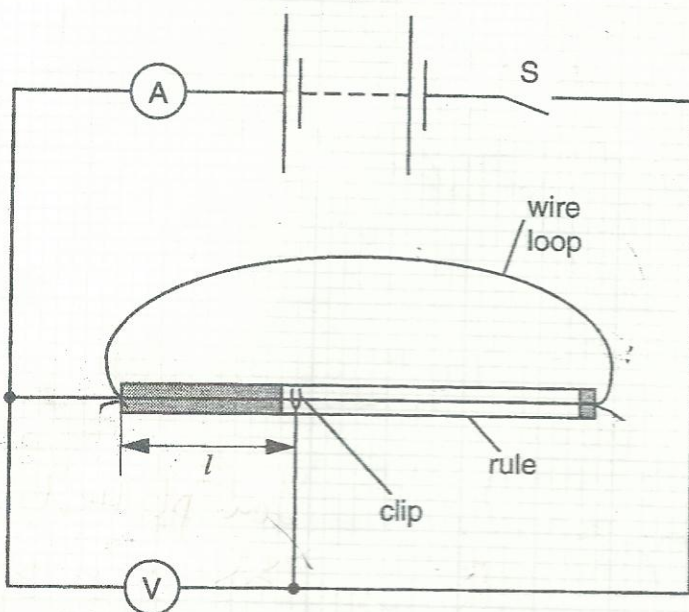


Fig. 2.1

- (b) (i) Place the clip on the wire so that the initial value of l is 0.900 m.
 (ii) Close switch S.
 (iii) For this value of l , measure the potential difference V and the current I . Record this set of readings.
 (iv) Move the clip to a different position on the wire and record the new values of V , I and l where l is in the range $0.35 \text{ m} < l < 0.90 \text{ m}$.
 (v) Repeat (iv) until you have six sets of values for V , I and l . Include in your table of results values of R/l , where $R = V/I$.
 (vi) Justify the number of significant figures which you have given for R/l .

- (c) R and l are related by the equation

$$\frac{R}{l} = k - \frac{kl}{L}$$

where L is the length of wire in the loop and k is a constant which depends upon the type of wire used.

- (i) Plot a graph of R/l (y-axis) against l (x-axis).
 - (ii) Determine the gradient and y-intercept of the graph.
 - (iii) Use your answers from (ii) to calculate values for L and k . Include appropriate units in each case.
- (d) 5.0 m of wire of the same diameter and material as that used in the experiment is made into a loop. Use the results of your experiment to calculate the resistance between two points on the wire which are 1.5 m apart.

DO NOT WRITE IN THIS SPACE

It is recommended that you spend about 30 minutes on this question.

- 3 Creep is the name given to the slow deformation of solid materials over an extended period of time when the material experiences stresses which are below that required to reach the elastic limit. An example of a situation where creep occurs is in the blades of a high temperature gas turbine. The operating temperature of the turbine is fairly close to the melting point of the material from which the blades are made. Therefore the blades are subject to creep and gradually become elongated as the turbine is used. This is illustrated in Fig. 3.1.

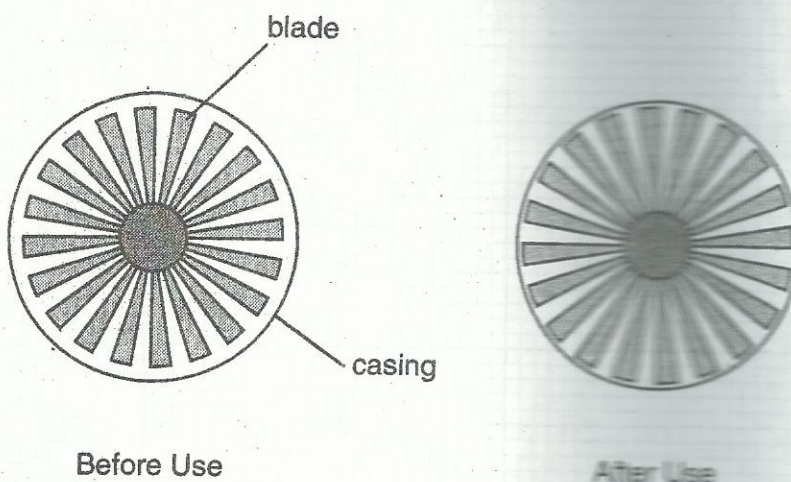


Fig. 3.1

The clearance between the blades and the casing is very small. This clearance decreases during the life of the turbine due to the creep in the blades. Therefore it is important to engineers to have information about the creep process so that the life-expectancy of the blades can be determined and damage to the engine can be prevented.

As a preliminary investigation of this effect, you are asked to design an experiment to show how the length of a wire made of lead changes with time (i.e. creeps) as the temperature of the wire and the load which it supports are changed.

You may assume that all of the equipment listed below is available, together with any other apparatus which is normally found in a school or college science laboratory.

Lead wire	Vernier scales
Microscope with scale	Bucket of sand
Thermometer	Long box with toughened glass sides
Stopwatch	Oscilloscope
Thermostat	Micrometer screw gauge
Set of masses	Electrical heater
Mass holder	Signal generator

You should draw a diagram showing the arrangement of your apparatus. In your account, you should pay particular attention to

- the procedure to be followed,
- the control of variables,
- any safety precautions which may be taken.

It is recommended that you spend about 30 minutes on this question.

4. A simple electromagnetic consists of a coil of wire mounted on a U-shaped soft iron core. The pole pieces P and Q are shaped as shown in Fig. 4.1 and are adjustable (i.e. the distance d may be altered).

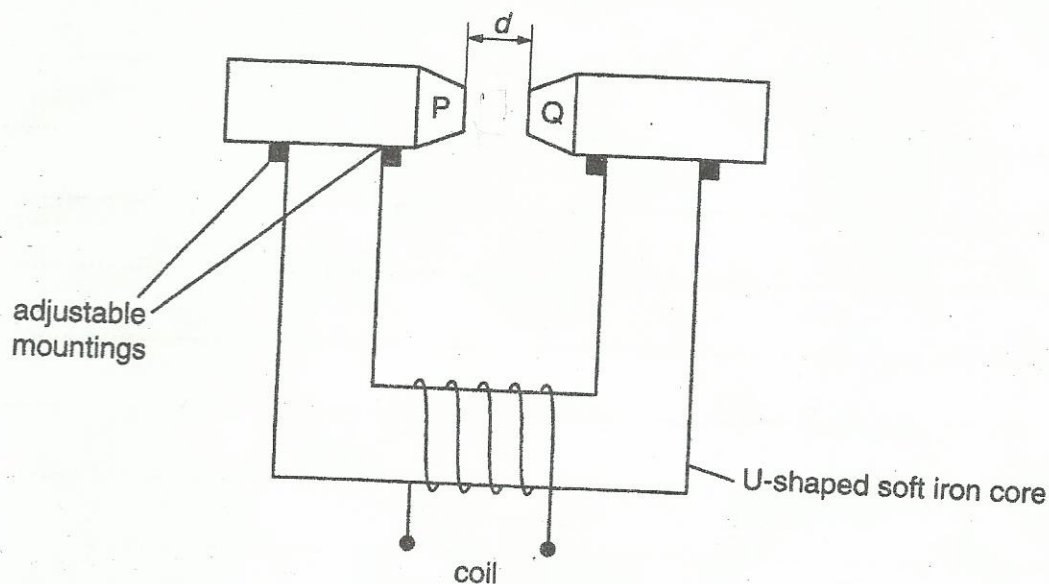


Fig. 4.1

When a current is passed through the coil a magnetic field is produced between P and Q.

Design an experiment to investigate how the magnetic flux density (field strength) between P and Q varies with

- (i) the current in the coil,
- (ii) the separation d of the pole pieces.

You should draw a diagram showing how the apparatus would be arranged. In your account, you should pay particular attention to

- (a) the equipment you would use for the investigation,
- (b) the procedure to be followed,
- (c) the control of variables.